Inclusive Reading Assistant

Milestone 1

Project name	Inclusive Reading Assistant
Principle / track	User Control and Agency
Mentored by	Mariam Mahmoud, Faculty of Computers and Artificial Intelligence
GitHub repo	https://github.com/Nesma-h/HCAI-Project
Target users	University students (for example with dyslexia, ADHD, or English as a second language)
Content	English academic text: lecture notes, textbook chapters, research papers

Idea for approval

The Inclusive Reading Assistant helps university students who struggle with dense English academic content. The user, not the system, decides how the content is adapted: they choose which support to use (simplify, summarize, explain, accessibility formatting), and they can always undo, adjust, or view the original text.

What makes it unique: a specific user group (university students with varied reading needs, without assuming everyone needs the same support), a specific content type (English academic text), and a design where nothing is simplified, summarized, or explained unless the student asks for it

Team members

Member	Role
Nesma Hesham	Frontend Developer
Menna Haitham	AI & Prompt Engineer
Malak Ahmed	NLP & Data Engineer
Jiovany Erwen	Testing & Evaluation Engineer
Mina Saber	Backend Developer

Problem statement

People who have difficulty reading or understanding complex content often struggle to access information independently. Long texts, difficult vocabulary, and complicated explanations can make it challenging for them to understand and use information in their daily lives or studies.

Our Inclusive Reading Assistant aims to make reading more accessible by allowing users to simplify text, summarize content, explain difficult words and concepts, translate text, or listen to it using text-to-speech. The assistant gives users control over how they want the content to be presented, allowing each person to choose the type of support that best fits their needs.

2. Development Workflow

From idea to chosen concept direction

	Phase	What we will do
1	**Problem Definition**	Identify the challenges university students face when reading complex English academic content
2	**Target User Identification**	Define university students as the target users and identify their reading needs and preferences
3	**Research & Requirements Analysis**	Research existing solutions and define the project's core features and requirements.
4	**Concept Generation**	Use two different AI tools to draft two interaction-flow concepts, compare them, and select a direction for approval
5	**Dataset Selection**	Research and select suitable English-language datasets for text simplification, summarization, and explanation.
6	**Data Preparation**	Explore, clean, and prepare the selected datasets for training or evaluation.
7	**Model Selection & Training**	Select suitable pretrained models, establish baselines, and fine-tune where necessary.
8	**Model Evaluation**	Evaluate output quality, readability, summary accuracy, and preservation of meaning
9	**System Design & Development**	Build the application interface and integrate the Simplify, Summarize, and Explain features with personalized controls.
10	**Testing & Improvement**	Test the system using academic content, identify issues, and improve the user experience and output quality.

3. Prompt

Same prompt, two tools: Claude and Gemini

Concept Task: Inclusive Academic Reading Assistant (Interaction Flow)

Project Overview Describe a conversational AI reading assistant (chatbot) for university students who need help understanding English academic content, such as lecture notes, textbook chapters, and research papers.

Track: User Control and Agency. The user, not the system, decides how the content is adapted and can always undo, adjust, or view the original.

Target users: University students who may struggle with dense academic text, for example due to dyslexia, ADHD, or English as a second language. Do not assume all students have the same needs.

Language scope: English only for both input and output.

Main Functions: Simplify (rewrite complex text in clearer English while preserving meaning). Summarize (condense into main ideas, only when the user asks). Explain (explain difficult concepts or terms based on the context of the sentence). Accessibility formatting (font, spacing, line length, background color). Users can select one function or combine several, and customize reading level, summary length, explanation depth, and formatting.

Your Task: Write ONE detailed concept of the core interaction flow, explaining in words how a university student uses the assistant from submitting academic content to receiving and refining the results. Explain: (1) how users provide content and choose functions; (2) how they customize preferences without unnecessary steps; (3) how results are presented and how users request changes, reject an output, or ask for further explanation; (4) how users compare output with the original and undo changes; (5) how empty or unsupported inputs and potentially inaccurate AI outputs are handled; (6) how the flow preserves user control, accessibility, and meaning.

Constraints: Text description only (no wireframes, mockups, UI layouts, diagrams, images, or code). Describe what the user does and what the chatbot says or does, not how the screen looks. Keep it simple and practical. Do not force users to use all functions. Explanations and summaries appear only when requested, or when the user turns on highlighting. Do not present AI-generated information as unquestionably correct. Do not assume a specific model, dataset, or technology. Do not invent statistics or results. State any assumption in one line.

Required Output: A. Concept Overview (distinctive name, 3-4 sentences). B. Core Interaction Flow (numbered steps: user actions, chatbot responses, controls). C. Example Scenario (a combination of functions and one rejected output). D. Strengths and Weaknesses (at least 3 each). E. Rationale (key decisions and assumptions; how they support User Control and Agency). Return ONE coherent concept, specific about user interactions. Keep the whole answer under 800 words.

Prompt log summary (Appendix A)

#	Tool	What it gave	Team decision
1	**Claude**	Draft 1: **ClearPath Reader**. 8-step flow, edge cases, scenario, 5 strengths, 5 weaknesses, rationale.	**Chosen** as the main direction (Page 6)
2	**Gemini**	Draft 2: **AdaptRead AI**. 6-step flow, scenario, 3 strengths, 3 weaknesses, rationale.	Not the base; some ideas reusable (Page 6)

5. Side-by-Side Drafts

Same prompt, two different AI tools

Section	Draft 1: Claude (ClearPath Reader)	Draft 2: Gemini (AdaptRead AI)
Assumption	Students have a text they are allowed to use; technology is undecided.	A flexible chat-and-panel environment (web or desktop) where text can be highlighted and transformed dynamically.
Concept	Chatbot where students paste or upload content and choose any, all, or none of the supports. Original always one tap away, every change can be undone, nothing explained or summarized unless asked. Output is a draft to check.	Student-driven reading companion and collaborative workspace. Users control reading difficulty, summary depth, and formatting; only requested functions run.
1. Input & functions	Paste, upload, or type. Chatbot confirms what it received and asks which function. No function is preselected; choosing nothing is valid.	Paste text and choose actions via control tags or natural prompts. Text loads into a split reading pane; only requested modules activate.
2. Preferences	At most one short question per chosen function, each with a default and a Skip. Formatting always available and instant. Saved settings never applied silently.	Sliders and toggles for reading level, summary length, and formatting, applied in real time without confirmation.
3. Results & refinement	AI-draft reminder on every output. Paragraph order kept. Optional term highlighting with on-demand explanation. Refine with "simpler", "keep technical terms", "shorter", or follow-ups.	Adapted text with inline context tags and a "Verify key terms" label. Users click Refine, Reject Output, or send a follow-up prompt.
4. Reject	"Not what I wanted" button, one optional question (too complex, too vague, changed the meaning), then retry or step back.	Reject Output / Reject Explanation, then a new prompt from the user.
5. Compare & undo	"Show original" toggle, sentence-by-sentence inspection, revert a sentence, paragraph, or everything, plus a version list.	"Compare Original" toggle (side-by-side or hover) and an Undo/Revert button for sentences or the whole text.
6. Edge cases	Messages for empty input and unsupported input (non-English, unreadable images, file types) with alternatives. Flags uncertain parts; keeps numbers, negations, and key terms unchanged.	Brief error for blank, non-English, or corrupted input. Low-confidence tag on jargon with a "Check Source Context" shortcut.
Scenario	Amira (ESL, psychology): Simplify + Explain, rejects a simplification that changed a term's meaning, retry, asks for a term explanation, checks the original, reverts one sentence, saves.	Maya (dyslexia, biology): Simplify + Explain Term, rejects a confusing explanation, asks for a truck analogy, saves.
Strengths	User picks functions; full reversibility; short skippable setup; uncertainty flags and visible original; meaning protection.	Independent function toggles; non-destructive editing; real-time accessibility formatting.
Weaknesses	Many controls may overwhelm; students may skip comparing; no guarantee for specialized text; sentence-level comparison takes effort; flags may be wrong.	Burden on the student to spot AI errors; context fragmentation; English-only limitation.

6. Comparison & Chosen Direction

Criterion	Draft 1: Claude	Draft 2: Gemini
Strengths	Very detailed flow with user action, chatbot response, and controls at each step. No preselected function, skippable questions, settings never applied silently. Full reversibility (reject, undo, version list). Detailed edge cases. Keeps numbers, negations, key terms. Scenario shows a combination, a rejection, and a revert.	Concise and easy to read. Good ideas: split reading pane, real-time sliders, "Check Source Context" shortcut. Simple scenario with a creative analogy refinement.
Weaknesses	Longer and more complex; many controls could overwhelm some students; sentence-level comparison adds effort.	Less detailed flow; no version history; rejection not a distinct step and no retry; fewer edge cases; some wording leans toward interface design; thinner rationale.
User control	Very strong at every step.	Good, but less complete.
Accessibility	Formatting available anytime; defaults and Skip reduce effort.	Strong real-time formatting; fewer options for varied needs.
AI errors	Draft label, uncertainty flags, key terms kept unchanged.	Draft label and low-confidence tag; fewer protections.

Chosen direction: Draft 1, Claude (ClearPath Reader)

We chose Draft 1 for these reasons:

1. **Best match with the track:** no preselected function, every question skippable, and the student can reject, refine, undo, or view the original at any point.
2. **More complete reversibility:** revert at sentence, paragraph, or whole-text level, plus a version list.
3. **Clearer rejection flow:** "Not what I wanted" asks why and retries.
4. **Better meaning protection:** every output is a labeled draft, uncertain parts are flagged, numbers, negations, and key terms stay unchanged.
5. **Better edge-case coverage:** empty, unsupported, unreadable, and non-English inputs each have a defined response.
6. **Closer to the prompt:** describes user and chatbot behavior in words, not interface layout.

Ideas to reuse from Draft 2: split reading pane, "Check Source Context" shortcut next to low-confidence flags, and real-time formatting. To avoid overwhelming users, keep defaults and Skip options visible and test the number of controls with students.